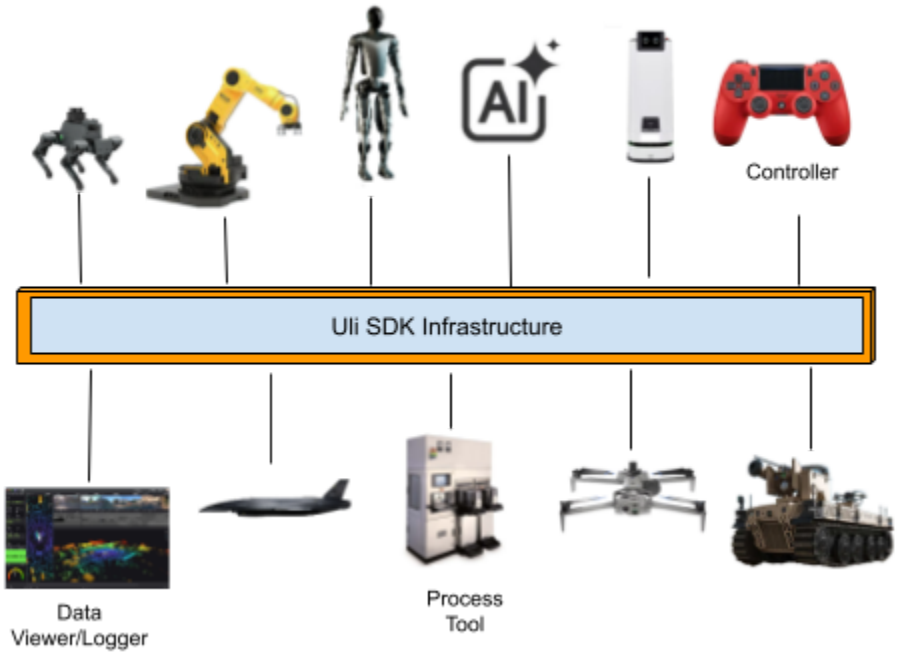
Bridging AI with Multi-Domain Capabilities

Uli (Unified Link Interface) SDK is the robotic nervous system for Agentic AI. It is a high-performance middleware designed to bridge the gap between traditional infrastructures and modern Agentic AI ecosystems. By leveraging the discovery services, Uli SDK enables physical assets to expose their functional capabilities and telemetry as discoverable “context” and “tools” for Language Models.

At the core of the Uli SDK's architecture is a **dynamic**, **self-configuring** infrastructure. This allows functional modules to be discovered, integrated, and utilized at runtime through a set of unified interfaces, eliminating the need for static, pre-configured connections and enabling true system agility.

This powerful infrastructure extends beyond individual modules to connect disparate systems—including **drones**, **autonomous vehicles**, **robot arms**, **process tools**, **controllers**, **AI agents**, and **data viewers/loggers**—that have been integrated using the Uli SDK. The same unified interfaces are leveraged to allow for the dynamic discovery of these complete systems and their functional capabilities and telemetry. This architectural approach is what facilitates true interoperability, enabling seamless collaboration between assets across multiple operational domains.

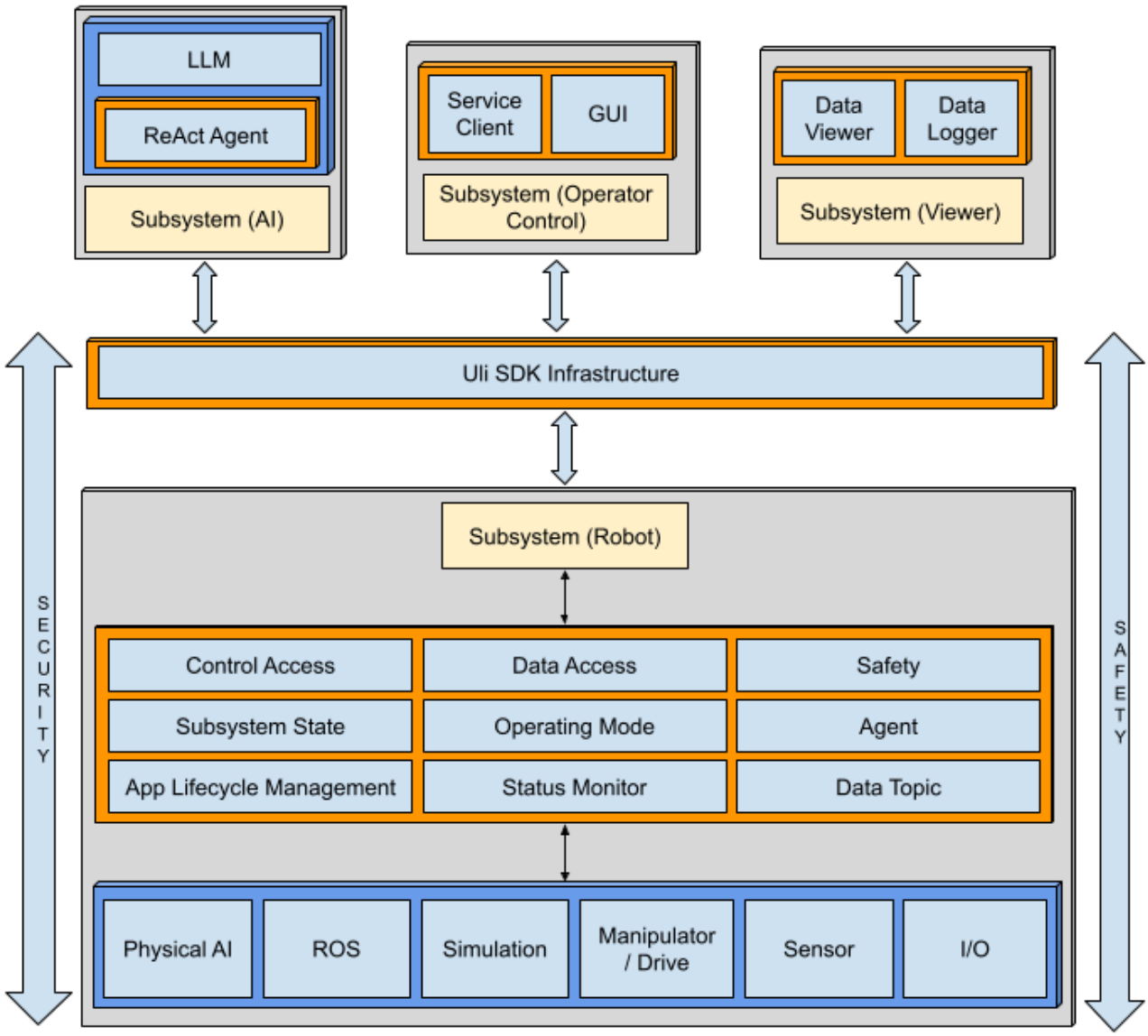


To accelerate development and ensure mission-readiness, Uli SDK provides a comprehensive suite of production-ready functional modules. This includes foundational pillars for the

Infrastructure, **Security**, **Safety**, and **Reliability**, which handle the most critical, non-negotiable aspects of robotic systems. In addition, the SDK offers application-level modules for **Operator Control**, **Data Viewing** and **Logging**, and **Agentic AI**, which can learn and act based on the context of the functional capabilities and telemetry exposed by the assets connected to the infrastructure.

The primary benefit of this modular, pre-built approach is that it empowers robotics developers to focus exclusively on their unique, value-adding logic—the core functional modules that differentiate their product. By providing a robust, pre-validated solution for the complex, non-differentiating aspects of development, the Uli SDK significantly shortens the **development lifecycle**, reduces **project risk**, and accelerates **time-to-market**.

Here is the high-level view of the Uli SDK architecture:



Uli SDK Implements Orange Blocks

Agile and Adaptable

Leveraging its DoD **MOSA**-compliant design with unified interfaces and dynamic modules, Uli SDK enables the seamless interoperability and rapid reconfiguration of capabilities, resulting in a highly agile system built for change.

Agent skills-based Middleware

Uli SDK serves as an implementation of **Agent Skills** for robotics. It acts as the bridge between an AI's **cognitive processes** and a **robotic** system's capabilities, enhancing situational awareness by feeding AI agents with the context of the robotic systems for reasoning and providing executable functional capabilities to act upon AI agents' decisions.

A2UI (Agent-to-UI) Framework

Uli SDK features native Dart-FFI integration, creating a powerful A2UI framework. This enables AI agents to drive real-time, high-fidelity user interfaces directly from telemetry streams.

Security

Uli SDK's security model uses certificate-based authentication and requires clients to make separate requests to gain explicit permissions for both **data access** (telemetry) and **control access** (functional capabilities).

Safety

Uli SDK's safety approach combines proactive detection with decisive handling through a redundant triggering system, which automatically responds to **application errors** and **manual e-stop commands** by transitioning the entire system into a coordinated emergency state.

Reliability

Uli SDK achieves reliability through a two-tiered **state machine system** that orchestrates both overall application and individual module behavior, combined with distinct operating modes that fine-tune functions for specific operational needs.

Integration

Uli SDK automatically generates Python bindings and Dart-FFI for applications, facilitating seamless integration with the **Physical AI**, **ROS**, **Simulations**, **AI Agents**, and a diverse array of **Python** and **Dart-Flutter** frameworks.

Development

Uli SDK develops applications for:

Devices – X86_64
Nvidia Jetson Nano, Xavier NX, Orin Nano, NX, AGX

Operating Systems – Ubuntu 18.04 - 24.04
Nvidia Jetpack 4.6, 5.1, 6.2

Tools

Uli SDK provides a comprehensive suite of tools to facilitate code generation, building, and application deployment. For code generation, the SDK's utilities are capable of producing concise C++ code for core components such as record structures, messages, services, and applications. It also automatically generates **Python bindings** and **Dart-FFI** to ensure seamless integration with various Python and Dart-Flutter frameworks. Via Dart-FFI, Uli SDK can serve as the back end for Flutter-UI for displaying media contents and 3D drawings. Additionally, developers can leverage the help from **Google Antigravity IDE** and **Claude CLI** that have comprehensive understanding of Uli SDK source codes and build systems, which helps speed up the development.

For the building process, the Uli SDK utilizes Google's highly efficient build tool, **Bazel**. This tool supports cross-building for multiple architectures, including both x86 and ARM 64-bit systems. To streamline the final phase of the development lifecycle, the SDK also includes shell scripts designed to simplify the staging and deployment of applications across a network of devices.

Highlights

1. **Agentic AI Interoperability** enabling autonomous agents to reason across diverse hardware fleets.

2. **A2UI (Agent-to-UI) Framework** native Dart-FFI integration, enabling AI agents to drive real-time, high fidelity user interfaces directly from telemetry streams.

3. **Dynamic and Self-Configuring** infrastructure for integrating robotic assets.

4. **Modular Open System Approach (MOSA)** provides interoperability and flexibility of the system architecture.

5. **Enhance LLM's situation awareness** from the connected ecosystem to power its cognitive reasoning and acting cycle.

6. **Certificate-based client** authentication and authorization.

7. **Exclusive control authorization** of agents, categorized into Operator, Maintainer, and Administrator.

8. **Data access authorization** categorizes into classified, controlled, and unclassified tiers.

9. **Lifecycle management** of the safety-critical applications, including startup, runtime, and shutdown phases.

10. **Service health reporting and monitoring** to track application status.

11. **Emergency stop (e-stop) propagation** to allow applications to manage critical situations.

12. **Support for various operating modes**, including Standard, Reduced, Training, Maintenance, and user-defined modes.

13. **Code generation tools** that produce C++ code for record structures, messages, services, applications, and Python bindings.

14. **Cross-build support** for x86_64, NVIDIA Jetson Nano, Xavier, and Orin, as well as multiple Ubuntu versions (18.04 - 24.04, Jetpack 4.6, 5.1, 6.2).

Advantages

Uli SDK is engineered to deliver a decisive edge in adaptability, interoperability, and operational integrity. Its architecture provides the following key advantages:

- **Agent Skills-based Middleware**

Implementation of the Agent Skills for robotics, ensuring zero-friction integration with the language models and agentic AI frameworks.

- **Self-Configuring and Adaptable Infrastructure**

Its dynamic, self-configuring infrastructure allows capabilities to be seamlessly added or removed, ensuring the system can rapidly evolve to meet new mission requirements.

- **Unified Discovery and Access**

A set of unified interfaces provides a consistent method for discovering and accessing the robotic assets and their functional capabilities and telemetry, dramatically accelerating development and integration.

- **Robust Security**

A robust, certificate-based authentication and authorization model enforces distinct permissions for data and control access to prevent misuse and protect critical assets.

- **Integrated Safety**

The integrated safety framework provides proactive monitoring of operator-initiated e-stop events and the health of critical applications to ensure a decisive response to emergencies.

- **Assured Reliability**

A hierarchical state management system, combined with distinct operating modes, provides granular control over application functions to ensure predictable and reliable system behavior.

- **High Quality**

Quality assurance is embedded throughout the development lifecycle, with continuous review and validation to ensure objectives are met.

- **Cost Effectiveness**

The SDK is delivered as open-source for licensed customers, offering full transparency and eliminating vendor lock-in to provide a low total cost of ownership.

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